

Shihang Liu

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Graduate Student
Department of Physics, Guangxi University
Supervisor: Dr. Bohua Li

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Research Interests 21cm Cosmology, First Galaxies, Dark Matter, Large Scale Structure Formation and Evolution

Education **Graduate Student Guangxi University (GXU)** Sept. 2023 – present
Major: Cosmology
Supervisor: Dr. Bohua Li

Undergraduate Yunnan University (YNU) Sept. 2018 – Jul. 2022
Elite Class for Astronomy
Major: Astronomy
Research Advisor: Prof. Xinzhong Er

Academic Activities 21cm Cosmology Summer School Jul. 2025
9th SKA Summer School Aug. 2024
“Love is the sky” summer camp, Xinjiang Astronomical Observatory Jul. 2021
Gravity meets Plasma workshop YNU Aug 2019

publication Li bohua, **Liu shihang**, Zhao menglin, et al.
Investigating the nature of dark matter and early-universe black hole formation based on the Hongmeng measurement of the global 21-cm spectrum [SSPMA](#)

Hayato Shimabukuro, **Shihang Liu**, Bohua Li.
Probing Fuzzy Dark Matter in the 21 cm Signal via Wavelet Scattering Transform. [arXiv](#)

Shihang Liu, Yilin Liu, Bowen Peng, Mengzhou Xie, Zelong Liu, Bohua Li, Yi Mao.
Constraining fuzzy dark matter with the 21-cm power spectrum from Cosmic Dawn and Reionization. [PRD](#), [arXiv](#)

Xinzhong Er, Jiangchuan Yu, Adam Rogers, **Shihang Liu**, Shude Mao,
Bias in apparent dispersion measure due to demagnification of plasma lensing on background radio sources. [MNRAS](#)

Projects	University students innovation and entrepreneurship project (YNU)	2019-2021
Award	First-class scholarship for graduate students (GXU)	2025
	First-class scholarship for graduate students (GXU)	2024
Skills	Programming and Professional Software • Python, C • 21cmFAST , 21cmSense	
Hobbies	Classical Music, Shell Collection, Animal Taxonomy	